Technical Project Report: Network Intrusion Detection & Threat Simulation

Executive Summary

The objective of this lab was to design, deploy, and validate an Intrusion Detection System (IDS) capable of identifying volumetric Denial of Service (DoS) attacks. The project focused on detecting TCP SYN Flood attacks while implementing thresholds to minimize false positives from legitimate high-traffic network events.

Lab Environment Architecture

Defender Architecture: Ubuntu Linux, Snort IDS (configured with Promiscuous Mode enabled)
Attacker Architecture: Kali Linux, hping3

TCP SYN Flood

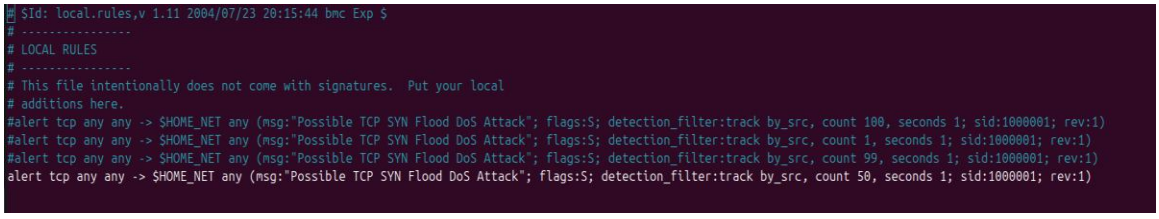
A TCP SYN flood is a type of Denial of Service (DoS) attack that exploits the standard TCP three-way handshake. The attacker sends a rapid succession of SYN (synchronize) requests to a target system. The target acknowledges these with SYN-ACKs and leaves the connections half-open, waiting for the final ACK. By flooding the server with these initial requests and never completing the handshake, the attacker exhausts the server's connection queues, rendering it unresponsive to legitimate traffic.

Detection Strategy & Signature Development

To detect this specific threat without overwhelming the monitoring infrastructure with false alerts during normal peak traffic, a custom Snort signature was authored and deployed to the local.rules configuration.

Custom Snort Configuration:

```
alert tcp any any -> $HOME_NET any (msg:"Possible TCP SYN Flood DoS Attack"; flags:S; detection_filter:track by_src, count 50, seconds 1; sid:1000001; rev:1)
```



Signature Component Breakdown:

- **alert tcp any any -> $HOME_NET any:** Monitors inbound TCP traffic originating from any external source traversing into the defined home network.
- **flags:S:** Isolates packets where only the SYN flag is set, indicating the initiation phase of the TCP handshake.
- **detection_filter:track by_src, count 50, seconds 1:** The critical thresholding mechanism. This instructs the IDS to track traffic statefully by source IP. An alert is only generated if a single source transmits 50 or more SYN packets within a 1-second window, distinguishing malicious volumetric floods from normal web scraping or database sync operations.

Attack Simulation & Validation

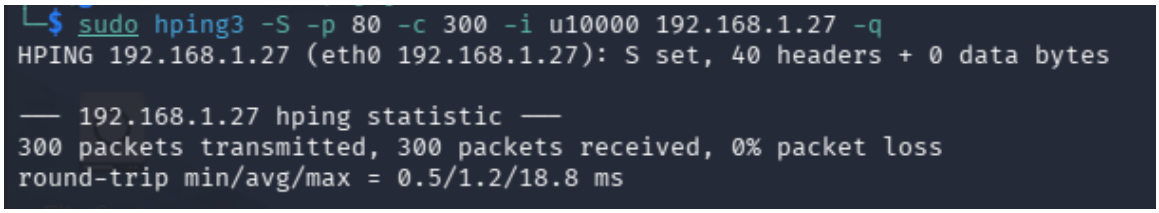
To validate the configuration, a SYN Flood attack was simulated using the hping3 packet crafting utility.

Attack Command:

```
sudo hping3 -S -p 80 -c 300 –i u10000 <192.168.1.27>
```

Command Flags:

- **-S:** Specifies the SYN flag on generated packets.
- **-p 80:** Sets destination port.
- **-c 300:** Sets the total number of packets to 300.
- **-i u10000:** Sets the time interval between each packet sent to 10,000 microseconds.



6. Results and Conclusion

Upon execution of the simulated flood, Snort successfully identified the volumetric anomaly and triggered the custom alert. The detection filter functioned as designed, proving that the monitoring environment was successfully configured to identify stateful anomalies based on defined threshold logic.

```
murewa@IndustryMachine:/etc/snort/rules$ sudo snort -A console -c /etc/snort/snort.conf -i wlp2s0 -q
09/02-00:30:48.859059  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2380 -> 192.168.1.27:80
09/02-00:30:48.869399  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2381 -> 192.168.1.27:80
09/02-00:30:48.879582  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2382 -> 192.168.1.27:80
09/02-00:30:48.889805  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2383 -> 192.168.1.27:80
09/02-00:30:48.900015  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2384 -> 192.168.1.27:80
09/02-00:30:48.910390  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2385 -> 192.168.1.27:80
09/02-00:30:48.920736  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2386 -> 192.168.1.27:80
09/02-00:30:48.931038  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2387 -> 192.168.1.27:80
09/02-00:30:48.941180  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2388 -> 192.168.1.27:80
09/02-00:30:48.951579  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2389 -> 192.168.1.27:80
09/02-00:30:48.961917  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2390 -> 192.168.1.27:80
09/02-00:30:48.972116  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2391 -> 192.168.1.27:80
09/02-00:30:48.983006  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2392 -> 192.168.1.27:80
09/02-00:30:48.993156  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2393 -> 192.168.1.27:80
09/02-00:30:49.003584  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2394 -> 192.168.1.27:80
09/02-00:30:49.013897  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2395 -> 192.168.1.27:80
09/02-00:30:49.024267  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2396 -> 192.168.1.27:80
09/02-00:30:49.035326  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2397 -> 192.168.1.27:80
09/02-00:30:49.045002  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2398 -> 192.168.1.27:80
09/02-00:30:49.055539  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2399 -> 192.168.1.27:80
09/02-00:30:49.065873  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2400 -> 192.168.1.27:80
09/02-00:30:49.076263  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2401 -> 192.168.1.27:80
09/02-00:30:49.086588  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2402 -> 192.168.1.27:80
09/02-00:30:49.096934  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2403 -> 192.168.1.27:80
09/02-00:30:49.109917  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2404 -> 192.168.1.27:80
09/02-00:30:49.120187  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2405 -> 192.168.1.27:80
09/02-00:30:49.130516  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2406 -> 192.168.1.27:80
09/02-00:30:49.140785  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2407 -> 192.168.1.27:80
09/02-00:30:49.151079  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2408 -> 192.168.1.27:80
09/02-00:30:49.161477  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2409 -> 192.168.1.27:80
09/02-00:30:49.171875  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2410 -> 192.168.1.27:80
09/02-00:30:49.182674  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2411 -> 192.168.1.27:80
09/02-00:30:49.193174  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2412 -> 192.168.1.27:80
09/02-00:30:49.203534  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2413 -> 192.168.1.27:80
09/02-00:30:49.213873  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2414 -> 192.168.1.27:80
09/02-00:30:49.224126  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2415 -> 192.168.1.27:80
09/02-00:30:49.234520  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2416 -> 192.168.1.27:80
09/02-00:30:49.244941  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2417 -> 192.168.1.27:80
09/02-00:30:49.255163  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2418 -> 192.168.1.27:80
09/02-00:30:49.265310  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2419 -> 192.168.1.27:80
09/02-00:30:49.275515  [**] [1:1000001:1] Possible TCP SYN Flood DoS Attack [**] [Priority: 0] {TCP} 192.168.1.21:2420 -> 192.168.1.27:80
```